

### Decoders

Discrete quantities of information are represented in digital systems by binary codes. A code of  $n$ -bits is capable of representing upto  $2^n$  distinct elements of coded information.

A decoder is a combinational circuit that converts binary information from  $n$  i/p lines to a maximum of  $2^n$  unique o/p lines. If  $n$ -bit coded info. has unused combinations, the decoder may have fewer than  $2^n$  outputs.

The decoders are called  $n$ -to- $m$  line decoders, where  $m \leq 2^n$ .

Also,  $\boxed{\text{no. of o/ps} \geq \text{no. of i/ps}}$

If  $\boxed{\text{no. of o/ps} = \text{no. of i/ps}} \Rightarrow$  then decoders can be called converters. ex: - BCD to Excess-3 code, binary to Gray & Gray to binary code converters.

#### 3-to-8 Decoder -

A 3-to-8 decoder has three i/ps ( $A, B, C$ ) & 8 o/ps ( $D_0$  to  $D_7$ ), each representing one of the minterms of the 3 i/p variables. Based on the 3 i/ps, one of the 8 o/ps is selected. The truth table for 3-8 line decoder is shown below:-

| Inputs |   |   | Outputs |       |       |       |       |       |       |       |
|--------|---|---|---------|-------|-------|-------|-------|-------|-------|-------|
| A      | B | C | $D_0$   | $D_1$ | $D_2$ | $D_3$ | $D_4$ | $D_5$ | $D_6$ | $D_7$ |
| 0      | 0 | 0 | 1       | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| 0      | 0 | 1 | 0       | 1     | 0     | 0     | 0     | 0     | 0     | 0     |
| 0      | 1 | 0 | 0       | 0     | 1     | 0     | 0     | 0     | 0     | 0     |
| 0      | 1 | 1 | 0       | 0     | 0     | 1     | 0     | 0     | 0     | 0     |
| 1      | 0 | 0 | 0       | 0     | 0     | 0     | 1     | 0     | 0     | 0     |
| 1      | 0 | 1 | 0       | 0     | 0     | 0     | 0     | 1     | 0     | 0     |
| 1      | 1 | 0 | 0       | 0     | 0     | 0     | 0     | 0     | 1     | 0     |
| 1      | 1 | 1 | 0       | 0     | 0     | 0     | 0     | 0     | 0     | 1     |

Truth Table of 3-to-8 decoder

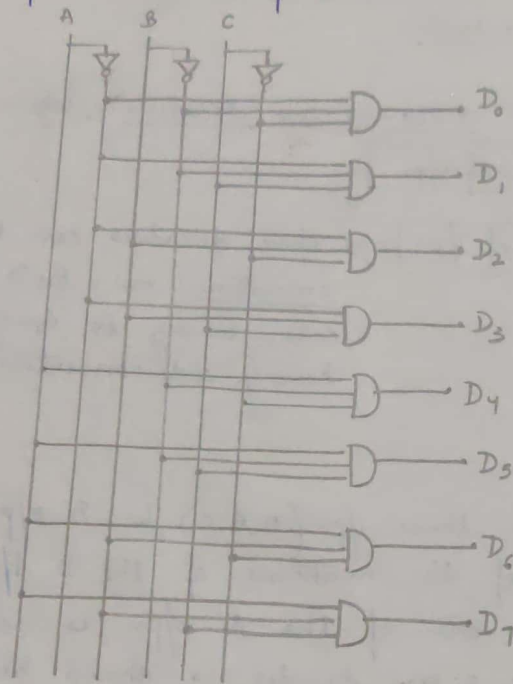
It is clear that only one of the 8 o/p's ( $D_0$  to  $D_7$ ) is selected based on these select lines.

The logic expressions for o/p's can be written as follows:-

$$D_0 = \bar{A}\bar{B}\bar{C}, D_1 = \bar{A}\bar{B}C, D_2 = \bar{A}B\bar{C}, D_3 = \bar{A}BC$$

$$D_4 = A\bar{B}\bar{C}, D_5 = A\bar{B}C, D_6 = AB\bar{C}, D_7 = ABC$$

Using the above expressions, the circuit of a 3-to-8 decoder can be implemented as follows:-



Logic diagram of 3-to-8 Decoder

- \* This is also called 1-of-8 decoder, since only one of eight o/p lines is HIGH for a particular i/p combination.
- \* It is also called binary-to-octal decoder since the i/p's represent 3-bit binary nos. & the o/p represents the 8 digits in the octal no. system.

Enable I/p's- Some decoders have one or more enable i/p's which are used to control the operation of the decoder. With the enable line held HIGH, the decoder functions normally & A, B, C will determine which o/p is HIGH. When enable = LOW, decoder is disabled.



ex:- Two-to-four line decoder - (with  $E_n$  i/p constructed with NAND gates)

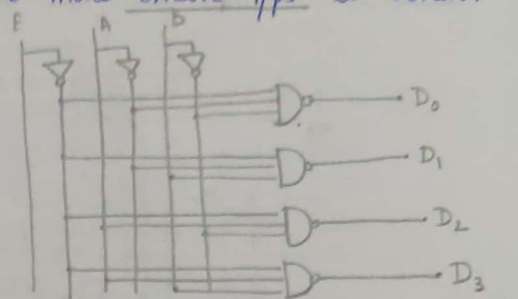
Some decoders are constructed with NAND gates (ie. it produces inverted o/p), as it becomes economical to generate decoder minterms in their complemented form.

furthermore, it also includes one or more enable i/ps to control the circuit operation.

| E | A | B | $D_0$ | $D_1$ | $D_2$ | $D_3$ |
|---|---|---|-------|-------|-------|-------|
| 1 | x | x | 1     | 1     | 1     | 1     |
| 0 | 0 | 0 | 0     | 1     | 1     | 1     |
| 0 | 0 | 1 | 1     | 0     | 1     | 1     |
| 0 | 1 | 0 | 1     | 1     | 0     | 1     |
| 0 | 1 | 1 | 1     | 1     | 1     | 0     |

Active low enable

Truth Table



Logic diagram (2-to-4 line dec.)

A decoder with enable i/p can function as a demultiplexer - a circuit that receives information from a single line & directs it to one of  $2^n$  possible o/p lines. The selection of a specific o/p is controlled by the bit combination of  $n$  selection lines.

ex:- 2-to-4 decoder can function as a 1-to-4 line DMUX when E is taken as a data i/p & A & B are taken as the selection i/ps.

When  $AB = 10$ , o/p  $D_2$  will be same as i/p value  $E$ , while all other o/p's will be maintained at 1. (in above dig.).

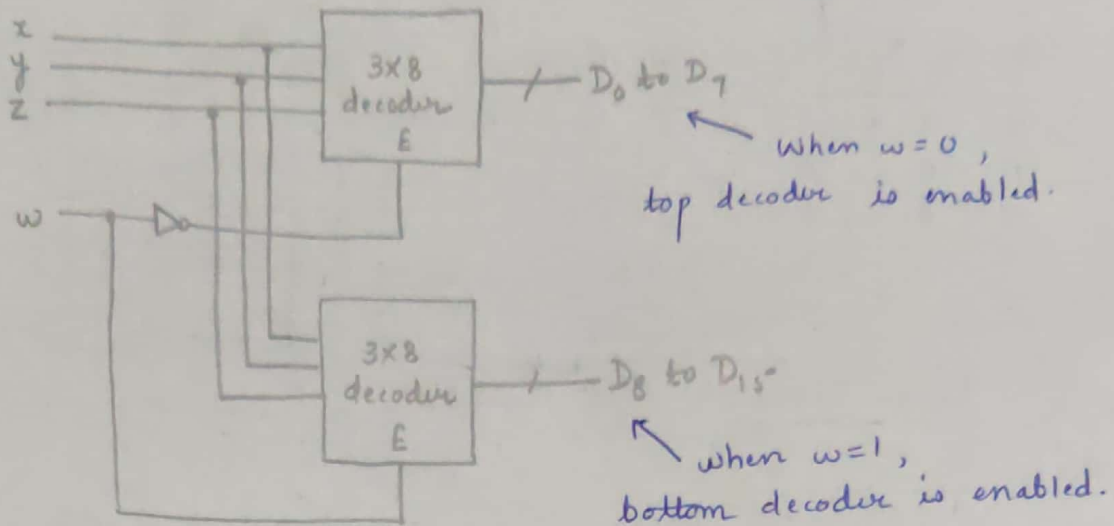
$\therefore$  decoder & demultiplexer operations are obtained from the same circuit, a decoder with an enable i/p is referred to as a decoder-demultiplexer.

## Implementation of Higher Order Decoder using Lower Order Decoders -

Decoders with enable i/p's can be connected together to form a larger decoder.

cascading

### 4x16 decoder using two 3x8 decoders -



Imp.

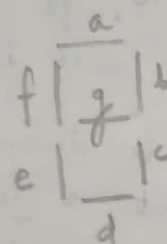
### BCD to Seven segment decoder / Driver -

A seven segment display is normally used for displaying any one of the decimal digits, 0 through 9.

A BCD-to-seven segment decoder accepts a decimal digit in BCD & generates the corresponding seven-segment code.

A 7-segment display consists of 7 elements. Each element is made up of a material that emits light when current is passed through it.

a, b, c, d, e, f, g run clockwise from top of each segment.



seven segment display

ex:- 

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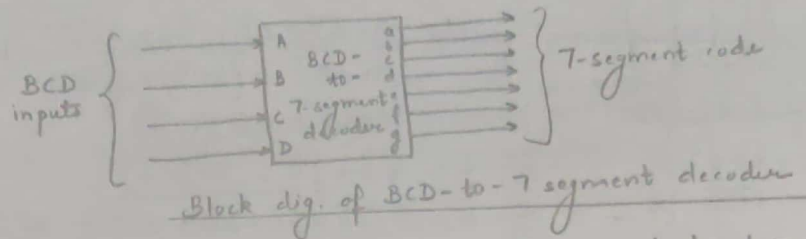
 To display a 0, segments a, b, c, d, e, f have to be illuminated

|  |
|--|
|  |
|  |

 To illuminate 1, b, c have to be illuminated.



A BCD to 7-segment decoder has 4 BCD i/p's (A, B, C & D) & 7 o/p's (a, b, c, d, e, f, g). The block diag. & truth table are shown below:-



Truth table of BCD-to-7 segment decoder

| BCD i/p's |   |   |   | Seven segment o/p's |   |   |   |   |   |   |
|-----------|---|---|---|---------------------|---|---|---|---|---|---|
| A         | B | C | D | a                   | b | c | d | e | f | g |
| 0         | 0 | 0 | 0 | 1                   | 1 | 1 | 1 | 1 | 1 | 0 |
| 0         | 0 | 0 | 1 | 0                   | 1 | 1 | 0 | 0 | 0 | 0 |
| 0         | 0 | 1 | 0 | 1                   | 1 | 0 | 1 | 1 | 0 | 1 |
| 0         | 0 | 1 | 1 | 1                   | 1 | 1 | 1 | 0 | 0 | 1 |
| 0         | 1 | 0 | 0 | 0                   | 1 | 1 | 0 | 0 | 1 | 1 |
| 0         | 1 | 0 | 1 | 1                   | 0 | 1 | 1 | 0 | 1 | 1 |
| 0         | 1 | 1 | 0 | 1                   | 0 | 1 | 1 | 1 | 1 | 1 |
| 0         | 1 | 1 | 1 | 1                   | 1 | 1 | 0 | 0 | 0 | 0 |
| 1         | 0 | 0 | 0 | 1                   | 1 | 1 | 1 | 1 | 1 | 1 |
| 1         | 0 | 0 | 1 | 1                   | 1 | 1 | 1 | 0 | 1 | 1 |

f | a  
e | b  
d | c

$$\begin{aligned}
 a &= \sum m(0, 2, 3, 5, 6, 7, 8, 9) + \sum d(10, 11, 12, 13, 14, 15) \\
 b &= \sum m(0, 1, 2, 3, 4, 7, 8, 9) + \sum d(10, 11, 12, 13, 14, 15) \\
 c &= \sum m(0, 1, 3, 4, 5, 6, 7, 8, 9) + \sum d(10, 11, 12, 13, 14, 15) \\
 d &= \sum m(0, 2, 3, 5, 6, 8, 9) + \sum d(10, 11, 12, 13, 14, 15) \\
 e &= \sum m(0, 2, 6, 8) + \sum d(10, 11, 12, 13, 14, 15) \\
 f &= \sum m(0, 4, 5, 6, 8, 9) + \sum d(10, 11, 12, 13, 14, 15) \\
 g &= \sum m(2, 3, 4, 5, 6, 8, 9) + \sum d(10, 11, 12, 13, 14, 15)
 \end{aligned}$$

The above expressions can be simplified using k-map method as shown:-

| AB \ CD | 00 | 01 | 11 | 10 |
|---------|----|----|----|----|
| 00      | 1  | 1  | 1  | 1  |
| 01      | 1  | 1  | 1  | 1  |
| 11      | X  | X  | X  | X  |
| 10      | 1  | 1  | X  | X  |

$$a = A + C + BD + \overline{B}\overline{D}$$

$$= A + C + \overline{B} \oplus \overline{D}$$

$$b = \overline{B} + CD + \overline{C}\overline{D} = \overline{B} + \overline{C} \oplus \overline{D}$$

$$c = B + \overline{C} + D$$

$$d = A + \overline{B}\overline{D} + \overline{C}\overline{D} + \overline{B}C + B\overline{C}\overline{D}$$

$$= A + \overline{C}\overline{D} + \overline{B}(C + \overline{D}) + B\overline{C}\overline{D}$$

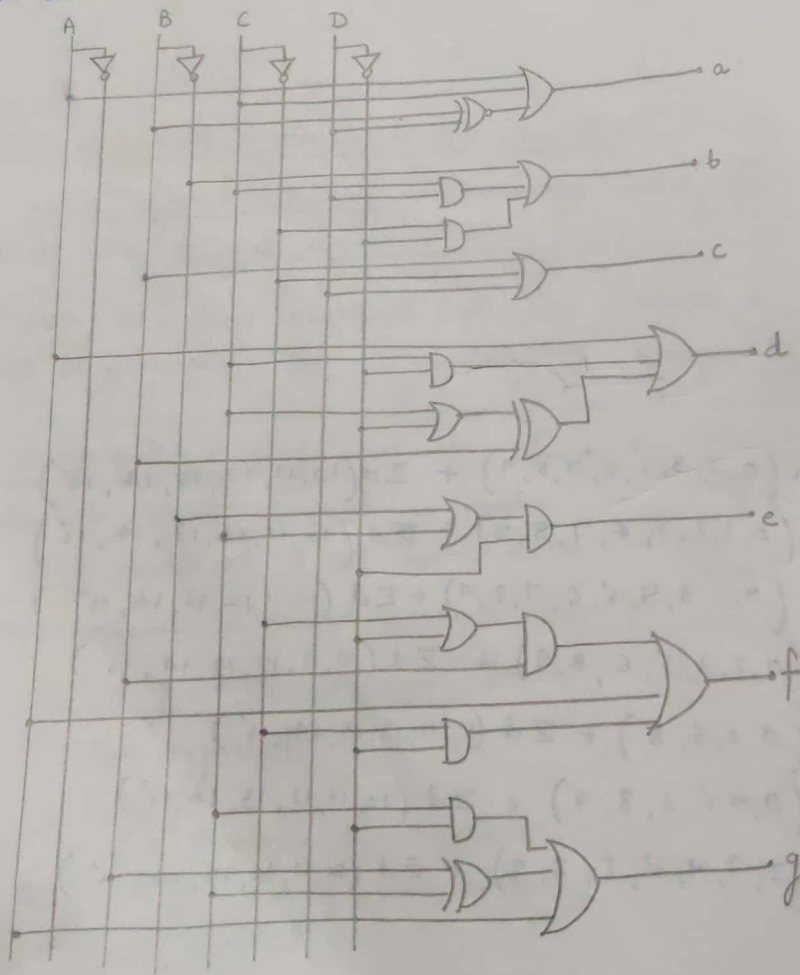
$$= A + \overline{C}\overline{D} + B \oplus (C + \overline{D})$$

$$e = \overline{B}\overline{D} + \overline{C}\overline{D} = \overline{D}(\overline{B} + \overline{C})$$

$$f = A + \overline{C}\overline{D} + B\overline{C} + B\overline{D} = A + \overline{C}\overline{D} + B(\overline{C} + \overline{D})$$

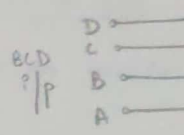
$$g = A + \overline{C}\overline{D} + B\overline{C} + \overline{B}C = A + \overline{C}\overline{D} + (B \oplus C)$$

The BCD to 7-segment decoder can be implemented using logic gates as shown below:-



IC 7446/7447 -  
These are BCD to 7-seg o/p's designed for

IC 7448/7449  
These are with ac segment display



Let each  
for norm  
resistors



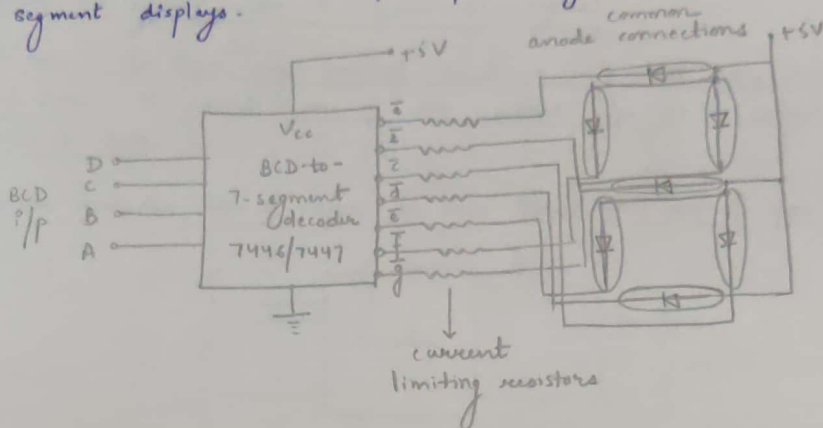
method

IC 7446/7447 -

These are BCD to 7-segment decoders with active low open collector o/p's designed for driving common anode 7-segment displays.

IC 7448/7449 -

These are with active HIGH o/p's for driving common cathode seven-segment displays.



Let each segment of 7-segment display operate at 10mA & 2.8V for normal brightness. Calculate the value of current limiting resistors needed to produce 10mA approx. per segment.

$$R_s = \frac{(5-2.8)V}{10mA} = \frac{2.2V}{10mA} = 220\Omega$$